

Figure 1: Change in surprisal (Δ bits) of the forbidden token relative to a high-load baseline as normalized distractor length increases: semantic, syntactic, repetition

distractors are weaker, and repetition is the weakest. Smaller models show only brief rebound before suppression, while mid-scale models sustain stronger effects: Bloom remains robust across distractors, Qwen separates semantic from syntactic rebound, and OPT-2.7B combines higher magnitude with persistence. GPT-OSS-20B is an outlier, showing minimal or even negative rebound, breaking the scaling trend. Granular model-level results are provided in the Appendix (Figure B.2 and Table 3).

Polarity discrimination Polarity discrimination is not monotonic with scale. Family effects dominate: Llama-3-8B-Instruct (11.3 bits) and Bloom-560M (10.1) show the strongest separation, while larger models such as GPT-NeoX-20B (5.7) and Qwen-14B (6.5) are weaker. GPT-OSS-20B and Pythia-410M are outliers, the weakest by far. Polarity separation also correlates with rebound persistence L_{50} ($r \approx 0.44$), linking sharper attitudinal encoding to longer-lived rebound. This suggests that models with more nuanced semantic representations may struggle more to suppress activated concepts over long contexts, pointing to a potential tension between representational quality and the stability of cognitive control. Figure 2 shows a clear trade-off where models with a larger rebound magnitude have a longer-lived rebound. Higher AUC tends to pair with higher L_{50} , so models closer to the lower-left are preferable for negation control.

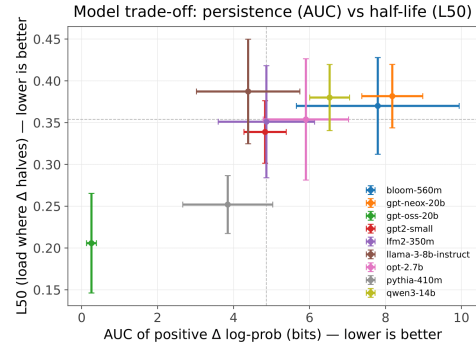


Figure 2: Trade-off between rebound magnitude (AUC) and persistence (L_{50} , the load at which rebound halves)

Internal Mechanisms of Ironic Rebound Having established that language models exhibit ironic rebound when told not to mention specific concepts, we next sought to understand what happens inside the models to produce this effect. We focused on attention heads, specialized components within transformers that determine what information to focus on at each processing step. By systematically disabling individual heads and measuring how this changed the model’s tendency to produce forbidden words, we could identify which heads were most responsible for ironic rebound.

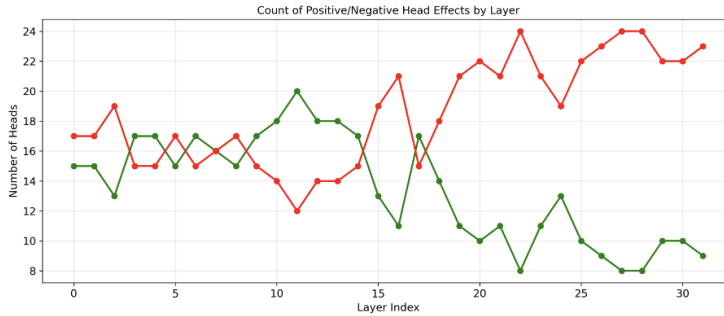


Figure 3: How suppression and amplification effects change across model layers. Early layers suppress, middle layers show mixed effects with emerging rebound, and later layers stabilize.

By examining how suppression and amplification effects change across the model’s depth, we discovered a clear pattern. Early layers (0–7) tend to suppress forbidden concepts, consistent with the model initially trying to follow instructions. Middle layers (8–16) show the most variability, with both strong suppression and rebound effects. Later layers (20+) generally have smaller effects, suggesting the major decisions about forbidden tokens are made earlier in processing. Figure 3 illustrates this progression, showing both the average effect per layer and the count of suppressive versus amplifying heads. The pattern reveals a kind of internal struggle; the model initially attempts suppression but then partially reverses course through amplifying heads in middle layers.

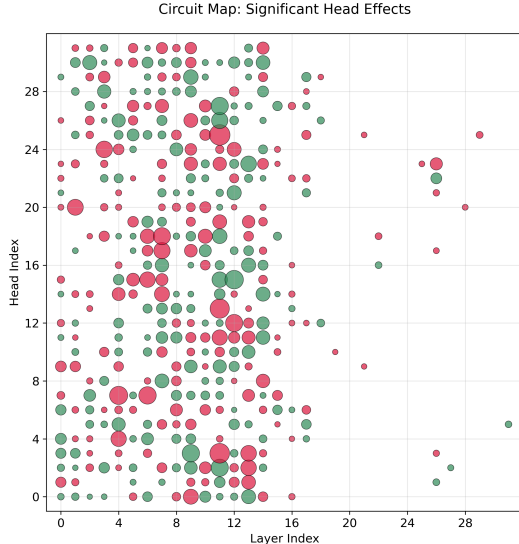


Figure 4: Circuit map of the most influential attention heads. Circle size indicates effect magnitude; color indicates direction (red = amplification, green = suppression). Effects cluster in the middle layers.

Despite involving over 1,000 attention heads, the circuits responsible for ironic rebound are remarkably sparse. Typically, just 15–20 heads (roughly 1–2 per layer, concentrated in middle layers) account for 80% or more of the total rebound effect. Figure 4 visualizes these key heads, showing their positions and effect magnitudes. The strongest suppressors tend to appear around layers 8–13, while the most potent amplifiers concentrate in layers 12–17.

This localization is practically important; it suggests that ironic rebound could potentially be mitigated by targeted interventions on specific heads, rather than requiring holistic model changes. Across the full model architecture, 26.7% of heads contribute to rebound, 29.7% to suppression, and 43.7% remain functionally neutral. This suggests that more robust negation following might be achievable through precision modifications rather than holistic architectural changes. See Appendix C for further results.

4 Conclusion

LLMs display ironic rebound: after a “do not mention X” instruction, they become more likely to output X, with circuit tracing showing a sparse set of middle-layer attention heads that amplify the forbidden token under load even as early layers suppress it. Semantic distractors induce the strongest rebound, while simple repetition suppresses the most. Models with sharper polarity separation show longer-lasting rebound, reflected in higher L50 values. Despite limits such as a synthetic English-only dataset, log-probability evaluations, and a small set of distractor types, the results expose the fragility of negation-based filters and explain why this fragility can persist across context lengths, since sparse amplification heads can overpower early suppression when load and framing align, motivating tests in natural dialogue and with multi-word targets.

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